

MUKESH.M

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Professional Summary

I am a passionate AI and ML Engineer with a strong interest in developing intelligent solutions that solve real-world problems. I enjoy working with Computer Vision, Deep Learning, Generative AI, and Autonomous Driving technologies, and I am constantly exploring emerging AI innovations to expand my knowledge. I am a quick learner, an enthusiastic problem solver, and someone who enjoys collaborating with teams to build scalable, impactful, and reliable AI-driven applications while continuously growing as an AI professional. I am highly motivated to embrace new challenges, contribute to innovative projects, and deliver efficient, high-quality solutions that create meaningful value for users and organizations.

Technical Skills

Programming	Python, SQL
Frameworks	PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, LangChain, LangGraph
Deep Learning	CNN, Attention U-Net, Transfer Learning, Semantic Segmentation, Object Detection
AI Concepts	Explainable AI (XAI), Feature Engineering, Data Augmentation, Model Optimization
GenAI / LLMs	RAG, Multi-Agent Systems, LLM Orchestration, Prompt Engineering, Qdrant
Backend & Web	FastAPI, React, PostgreSQL, Docker
Tools	Git, GitHub, ONNX, CARLA Simulator, MySQL Workbench, Tableau, Power BI

Projects

Attention U-Net based Brain Tumor Segmentation using Explainable AI:

Developed an automated brain tumor segmentation framework using Attention U-Net on MRI scans. Achieved **98.7% Dice Score** and **97.9% IoU**, outperforming traditional segmentation approaches. Integrated attention gates and skip connections to improve tumor localization and preserve spatial features. Implemented Grad-CAM based Explainable AI visualization for interpretable clinical decision support. Technologies Used: PyTorch, OpenCV, Medical Image Processing, Deep Learning.

Lung Disease Detection and Recommendation System using Explainable AI with Metadata:

Developed a CNN-based chest X-ray classification model with metadata fusion for lung disease detection. Achieved **96% accuracy** and **0.93 F1-score** on 5,000+ labelled X-ray images. Optimized inference using ONNX for fast CPU deployment and implemented Grad-CAM visualization. Integrated a rule-based recommendation engine for symptom analysis and clinical support. NIH ChestX-ray14 and Kaggle Chest X-ray datasets. (Attended IEEE International conference and published in Scopus index: Lung Disease Detection and Recommendation System using Explainable AI with Metadata | IEEE Conference Publication | IEEE Xplore. DOI: [10.1109/ICMLAS67792.2026.11483688](https://doi.org/10.1109/ICMLAS67792.2026.11483688))

AI-Powered ADAS Copilot – Real-Time Perception, Risk Prediction & LLM-Based Driver Assistance:

Built a real-time Advanced Driver Assistance System combining **object detection**, **lane detection**, and **monocular depth estimation** to perceive the driving environment from live video. Designed a collision risk-prediction and tracking engine fusing detection, tracking, and depth outputs to flag high-risk events, validated in CARLA Simulator. Layered an LLM-based copilot on the risk engine to generate natural-language driving alerts on a live dashboard. Technologies Used: Python, OpenCV, YOLO, CARLA Simulator, LLM Agents.

AutoMind AI – Multi-Agent GenAI Workspace for Automotive Engineering:

Developed a multi-agent Generative AI workspace that helps automotive engineers analyze technical documents, understand vehicle diagnostics, and retrieve engineering knowledge using **Retrieval-Augmented Generation (RAG)** and AI agents. Built a FastAPI backend orchestrated with LangChain/LangGraph and a Qdrant vector database, paired with a React + TypeScript frontend, containerized with Docker and backed by PostgreSQL. Technologies Used: Python, FastAPI, LangChain, LangGraph, Qdrant, React, TypeScript, PostgreSQL, Docker.

Education

Vellore Institute of Technology (VIT)

2024 – 2026

M.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning) **CGPA: 8.22/10**

KINGSTON Engineering College

2019 – 2023

Bachelor of Engineering in Mechanical Engineering

(Relevant Coursework: AutoCAD, Nx-CAD, SolidWorks, Catia and Ansys) **CGPA: 8.9/10**

Certifications

- Data Science & Artificial Intelligence Certification | I.T. Vedant** (Jan 2024 – June 2024 | 6 Months)
Details: Mastered Data Analytics, SQL, Python, ML, Tableau, Power BI, and AI techniques including Deep Learning, NLP, CV, and Time Series Analysis.
- Fundamentals of Machine Learning with AI – Hands-on ML model development and evaluation.
- IBM DA0101EN – hands on experience in Data Analytics in python